RISCFORGE 1TOPS

IP Verification Status Sheet

Date: 24 June 2026 — **Team:** Verification — **Target:** 22nm

Master Progress

#	IP	Function / Role	Bus	Verification	RTL	PD	Notes / Blockers
1	**APB SPI Master**	Boot-time instruction fetch from external SPI flash	APB	✓ Functionally Verified	✓	✓	—
2	**AXI-Lite → APB**	Protocol conversion: crossbar → peripheral bus	AXI-L / APB	○ Not Started	○	○	Needed before peripheral integration
3	**GPIO**	General-purpose I/O, LED/switch/sensor control	Reg IF	○ Not Started	○	○	Confirm GPIO count & clock-gate variant
4	**CLIC**	Core-local interrupt controller	Reg IF	○ Not Started	○	○	Requires modified CV32E40P
5	**APB Timer**	OS tick / delay generation via compare-match IRQ	APB	○ Not Started	○	○	—
6	**APB I2C**	Sensor / EEPROM comms (I2C master)	APB	○ Not Started	○	○	Open-drain pads required for SCL/SDA
7	**APB UART**	Serial debug / console communication	APB	♦ In Progress	○	○	TB covers TX/RX, loopback, FIFO, interrupts

Legend & Definitions

Status: ✓ Complete ♦ In Progress ◯ Not Started

IP Name — The specific peripheral block (linked to RTL repository).
Function/Role — Brief one-liner on the block's primary purpose.
Verification Status — Not Started / In Progress / Functionally Verified / Signed Off.
PD Notes — Physical Design dependencies, padding, or clocking considerations.

Per-IP Quick Reference

1. APB SPI Master ✓ Verified

Item	Detail
Function	Full-duplex SPI master with TX/RX FIFOs; used for boot-time instruction loading from external SPI flash.
Bus	APB slave
Key Ports	`spi_clk`, `spi_csn[0:3]`, `spi_sdo[0:3]`, `spi_sdi[0:3]`, `events_o[1:0]`
Source	https://github.com/pulp-platform/apb_spi_master
Verification	✓ **Fully verified:** 14-test BFM-based suite — register RW, FIFO depth, SPI modes, loopback, RISC-V instruction injection.
PD Notes	Top-level SPI pins + APB interface already identified; clock-domain: single `HCLK`.

2. AXI-Lite → APB Bridge ◯ Not Started

Item	Detail
Function	Protocol converter: translates AXI-Lite transactions from the crossbar into APB transactions for peripherals.
Bus	AXI-Lite slave → APB master
Key Ports	Full AXI-Lite slave (AW/W/B/AR/R channels), APB master (`PADDR`, `PWDATA`, `PRDATA`, `PSEL`, `PENABLE`, `PWRITE`, `PREADY`, `PSLVERR`).
Source	https://github.com/pulp-platform/axi/blob/master/src/axi_lite_to_apb.sv
Verification	◯ **Not verified:** Single-beat read/write, back-to-back, error propagation.
PD Notes	Pure combinational/registered logic, no memories; single file.

3. GPIO (PULP) ◯ Not Started

Item	Detail
Function	64-GPIO controller with push-pull / open-drain modes, 2-stage input sync, per-pin edge & level interrupts.
Bus	Register Interface (APB/AXI-Lite wrappers available)
Key Ports	`gpio_in`, `gpio_out`, `gpio_tx_en_o`, `global_interrupt_o`, `pin_level_interrupts_o`
Source	https://github.com/pulp-platform/gpio
Verification	◯ **Not verified:** Input sync, output modes, interrupt types (rise/fall/high/low), GPIO enable/disable.
RTL Notes	Has clock-gating variant (`gpio_with_clk_gates`) — confirm target with PD; GPIO count configurable.
PD Notes	Per-pin clock gates may affect CTS; `gpio_tx_en_o` drives pad OE.

4. CLIC (PULP)

◯ **Not Started**

Item	Detail
Function	Core-Local Interrupt Controller — pre-emptive, vectored, priority/level-based interrupts for RISC-V core; replaces basic CLINT.
Bus	Register Interface (APB wrapper available)
Key Ports	Interrupt request lines to core, per-interrupt config registers, `cliccfg`, `clicintctl`
Source	https://github.com/pulp-platform/clic
Verification	◯ **Not verified:** Priority resolution, level thresholding, vectored dispatch, edge vs level triggering.
RTL Notes	Requires modified CV32E40P core; up to 4096 interrupt lines (default 256).
PD Notes	Parameterised width — confirm `NUM_INTERRUPT` and `CLICINTCTLBITS` before synthesis.

5. APB Timer (PULP)

○ **Not Started**

Item	Detail
Function	Simple 32-bit timer(s) with compare-match interrupt via APB; used for OS tick / delay.
Bus	APB slave
Key Ports	PADDR, PWDATA, PRDATA, APB control, irq_o
Source	https://github.com/pulp-platform/apb_timer
Registers	Timer value, timer compare, timer control (per timer instance)
Verification	○ **Not verified:** Timer count, compare-match IRQ, prescaler, enable/disable.
PD Notes	Small footprint, no memories.

6. APB I2C (PULP)

○ **Not Started**

Item	Detail
Function	I2C master controller with bit-level & byte-level FSMs, clock prescaler, arbitration loss detection.
Bus	APB slave
Key Ports	scl_pad_i/o, scl_padoen_o, sda_pad_i/o, sda_padoen_o, interrupt_o
Source	https://github.com/pulp-platform/apb_i2c/blob/master/apb_i2c.sv
Registers	CLK_PRESCALER, CTRL, RX, STATUS, TX, CMD
Verification	○ **Not verified:** Byte read/write, ACK/NACK, arbitration loss, clock stretching, interrupt generation.
PD Notes	Open-drain pads required for SCL/SDA (_padoen_o active-low).

7. APB UART

♦ **In Progress**

Item	Detail
Function	Serial debug / console communication (16550-compatible).
Bus	APB slave
Key Ports	tx, rx, APB bus interface, interrupt_o
Source	https://github.com/pulp-platform/apb_uart
Verification	♦ **Partial verification started:** 15 TCs + SVA written, sim pending. TB covers TX/RX, loopback, FIFO, interrupts, modem, reset.
PD Notes	Minimal gating required. Standard pads sufficient.